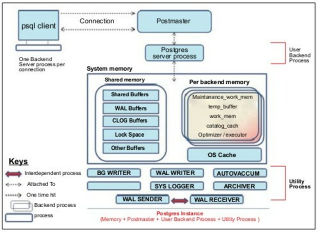

Figure 1: Components of a PostgreSQL database

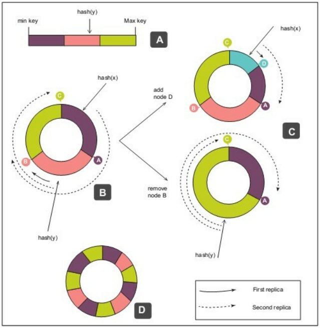

Figure 2: Consistent hashing

Data in motion

How data is received, how it is processed and then consumed within the context of its spatial temporal attributes, is beginning to determine the choice of databases.

There is also a subtle shift in terminology from databases to the notion of a *data store*.

A *data store* allows distribution of data across servers. This has two benefits: (a) scalability, and (b) improved performance for both ‘updates’ and ‘queries’.

Consistent hashing is a promising technique, which is used to implement data distribution and introduces the notion of a *key* element. In the context of a relational database, this is the primary key and, in a data store, this is just the key.

The most important characteristic is that there can only be a *single entry* for the key in a data store. This single entry is assigned to a numerical value, which is assigned to a specific server. Thus, the specific key is always assigned to the same server for storage and querying.

The advantage is that the load can be distributed across the servers at the cost of reliability. If any of the servers crash, the data stored on that server becomes unavailable. With an increasing number of servers, the probability of a crash keeps going up.

To improve reliability, we apply the *consistent hashing* technique.

The data stores that implement this approach are:

- Amazon Dynamo
- Memcached
- Project Voldemort
- Apache Cassandra
- Skylable
- Akka’s hashing router
- Riak, a distributed key value database

RDF (Resource Description Framework)

The RDF data model is based on the idea of making statements about resources in the form of subject-predicate-object expressions (triples).

Subject is the resource. *Predicate* refers to the traits of the resource and expresses a relationship between the subject and the object.

RDF is an abstract model with several serialisation formats like Turtle, JSON-LD and RDF/XML.

For an example, look at the following sample RSS 1.0 blog feed:

```
<?xml version="1.0"?>

<rdf:RDF
  xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
  xmlns="http://purl.org/rss/1.0/">

  <channel rdf:about="http://www.xml.com/xml/news.rss">
    <title>XML.com</title>
    <link>http://xml.com/pub</link>
    <description>
      XML.com features a rich mix of information and services
for the XML community.
    </description>

    <image rdf:resource="http://xml.com/universal/images/
xml_tiny.gif" />

    <items>
      <rdf:Seq>
        <rdf:li resource="http://xml.com/pub/2000/08/09/xslt/
xslt.html" />
        <rdf:li resource="http://xml.com/pub/2000/08/09/
rdfdb/index.html" />
      </rdf:Seq>
    </items>
```